

CoRoT-2b

R-0007

Benchmark run · workflow W8-benchmark · record 2026-bench_CoRoT-2_260630 · created 2026-07-08T03:30:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461221.7291 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.019
Transit depth (Rp/Rs)^2	3.9e-07 +/- 0.00023 [%]
Orbital Inclination (inc)	87.68 +/- 2.07
Airmass coefficient 1 (a1)	1.261 +/- 0.04
Airmass coefficient 2 (a2)	-0.1458 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	3.2 %
Best Comparison Star	None
Optimal Aperture	7.29
Optimal Annulus	8.66
Transit Duration (day)	0.0783 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.019 vs 0.1667 (deviation 3.96 σ)
Duration fit vs published	0.0783 d vs 0.095 d (deviation 0.94 σ)
Mid-time O-C	5.3 min
Depth SNR	0.0
Residual scatter	3.2 %

Light curve

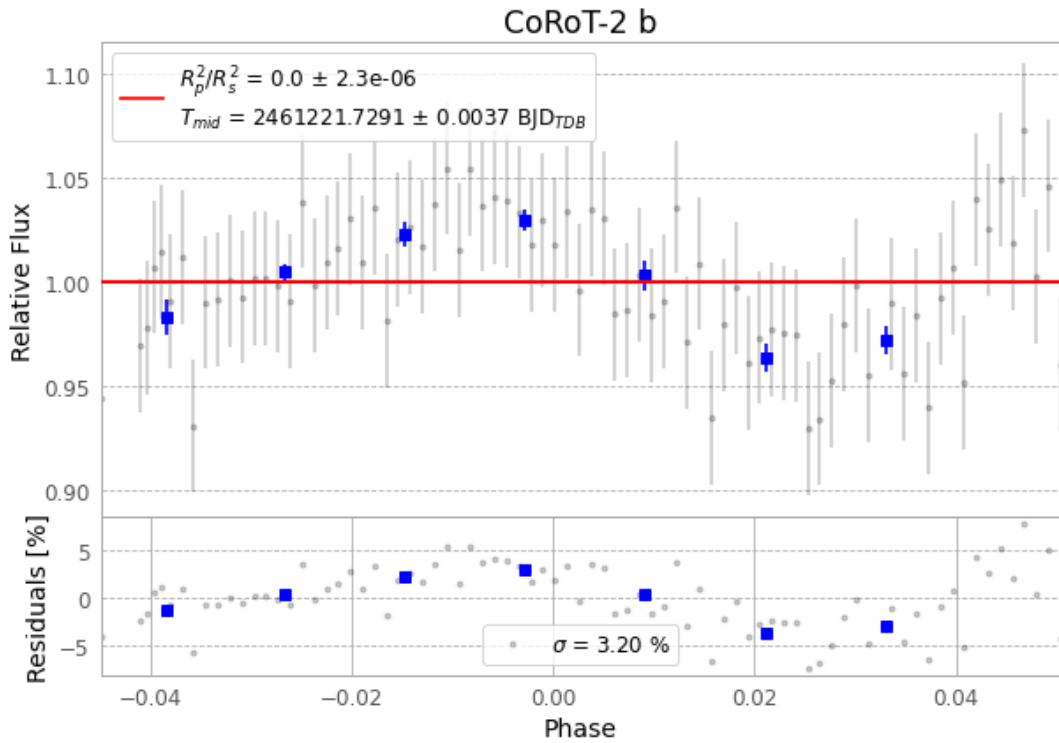


Figure 1 — FinalLightCurve_CoRoT-2 b_2026-06-29.png (SHA-256 0c41b58a20a8d985...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [276, 189], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f5da2df7666cd118...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

81 FITS frames (53 MB), CoRoT-2260630032805.FITS → CoRoT-2260630072716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-260630.log (7d5cd08fd169...), FinalLightCurve_CoRoT-2 b_2026-06-29.png (0c41b58a20a8...), FinalLightCurve_CoRoT-2 b_2026-06-29.csv (ee58d446eebd...)

Compute resources

Wall time 244.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.